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Occupancy-Aware Smart Library Seat Reservation with Predictive Demand Analytics

CSA0925-Java programming

Under the Guidness of
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Problem Background

Application Domain

- Smart Library Management
- Artificial Intelligence (AI)
- Data Analytics

Current Scenario

- Students spend considerable time searching for vacant seats.
- Libraries often experience overcrowding during examinations.
- Manual seat allocation causes confusion and inefficient utilization.
- No accurate prediction of future seat availability.

Need for the Project

- Efficient seat utilization.
- Reduce waiting time.
- Improve student convenience.
- Assist library administrators in resource planning.

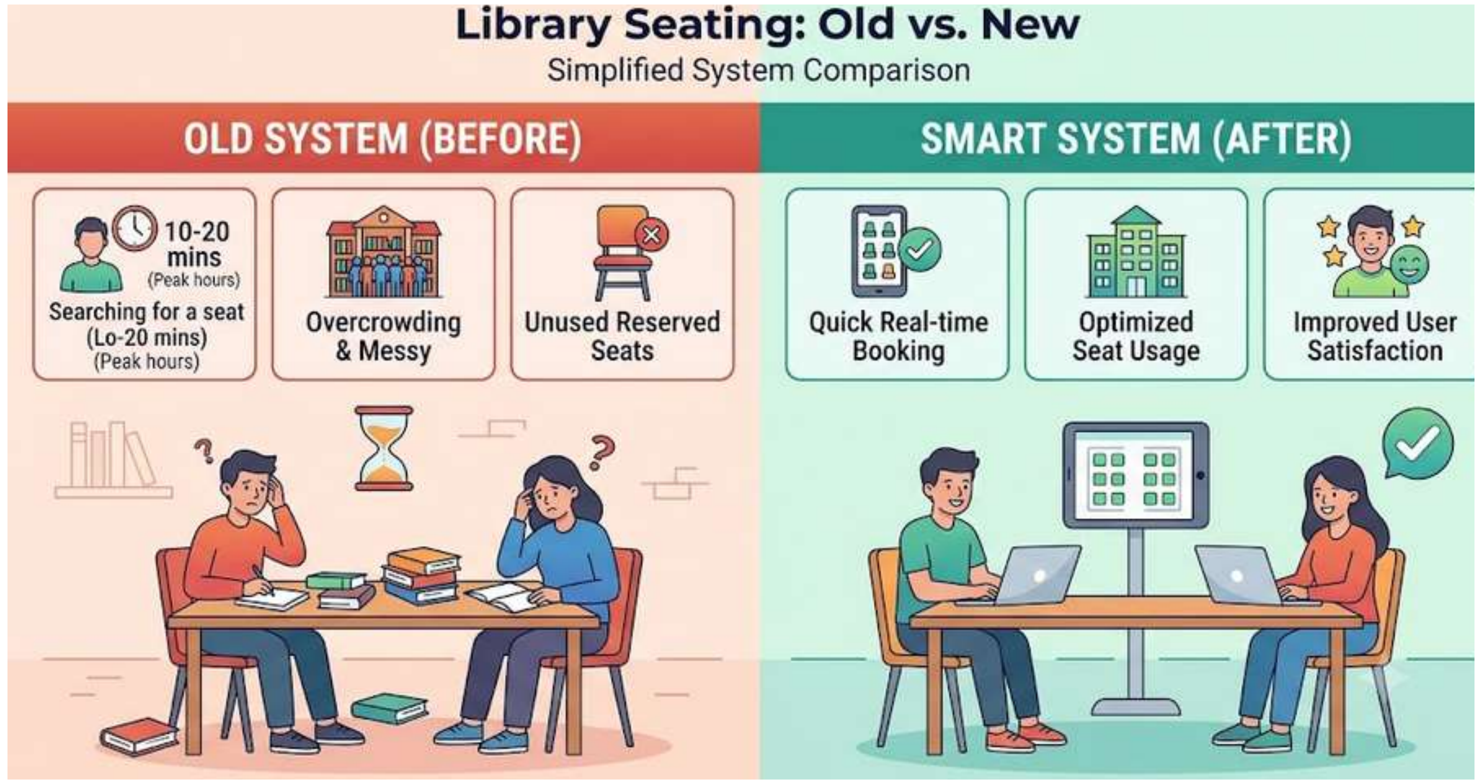
Target Users

- Students
- Faculty Members
- Library Staff
- Library Administrators

Problem Statement

Problem Definition

Traditional library seating systems lack real-time occupancy monitoring and reservation facilities, causing unnecessary waiting, inefficient seat usage, and overcrowding.



Evidence / Statistics

- Students spend 10–20 minutes searching for seats during peak hours.
- Library occupancy exceeds 90% during examinations.
- Around 20–30% of reserved seats remain unused due to no-shows.

Limitations of Current Systems

- No live seat availability.
- Manual reservation process.
- No occupancy prediction.
- No automatic cancellation of unused reservations.

Expected Impact

- Faster seat allocation.
- Better utilization of available seats.
- Reduced overcrowding.
- Improved user satisfaction.

Objectives

Primary Objective

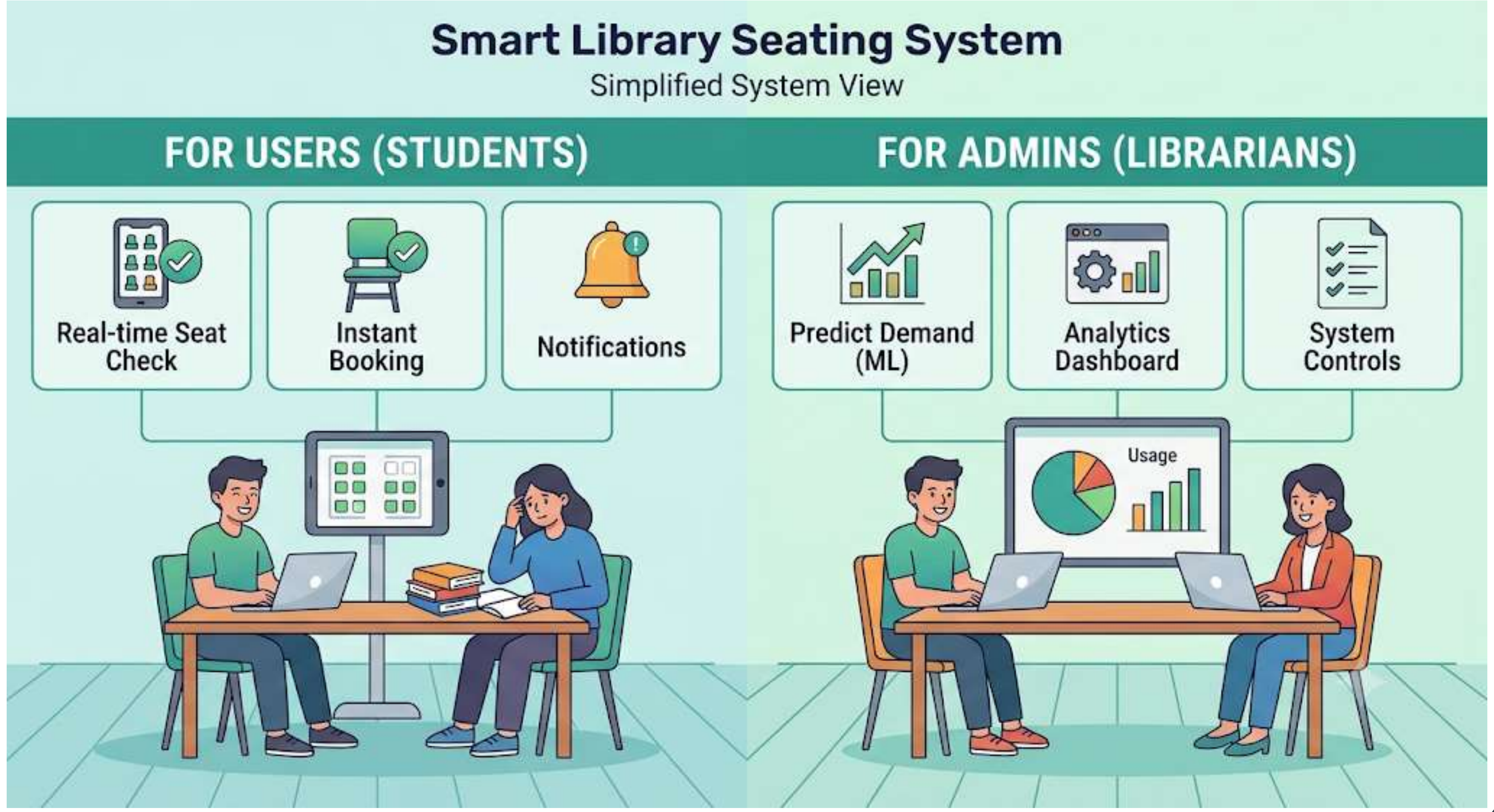
Develop an intelligent library seat reservation system that provides real-time occupancy monitoring and predicts future seat demand.

Project Scope

- Smart seat booking
- Occupancy monitoring
- Predictive analytics
- Dashboard for administrators
- Mobile/Web application

Measurable Objectives

1. Display real-time seat availability.
2. Enable online seat reservation.
3. Predict occupancy using Machine Learning.
4. Send reservation notifications.
5. Generate analytics for administrators.



LITERATURE SURVEY

Research PaperKey	Year	Contribution
Enabled Smart Library Seat Management for Improved Resource Allocation	2025	Developed an real-time seat monitoring and efficient resource allocation.
Optimizing Smart Library Spaces: Integrating PIR Sensors, Credit-Based Booking Systems, and Advanced Algorithms for Efficient Resource Management and Space Allocation	2025	Used PIR sensors, seat booking, and Random Forest prediction to optimize library space utilization.
A Systematic Review Smart Technologies for Seat Occupancy and Reservation Needs in Smart Libraries	2023	Reviewed, reservation systems, and identified research gaps in smart library management.
Power Management in Smart Residential Building with Deep Learning Model for Occupancy Detection	2022	Demonstrated deep learning-based occupancy detection methods that can be adapted for smart seating systems.
Smart Library Occupancy Management System using Deep Learning	2024	Implements a deep learning-based system (YOLOv8, YuNet, MobileFaceNet) for real-time occupancy detection and reservation management.

Research Gap

- Existing systems:
- Only monitor occupancy
- No demand prediction
- No smart reservation
- Limited analytics

Proposed work combines

- AI prediction
- Reservation
- Analytics Dashboard

Existing vs proposed

Existing System	Proposed System
Manual seat search	Automatic seat booking
No live tracking	Real-time occupancy
No prediction	Demand forecasting
No notifications	Reservation alerts
Static reports	Interactive dashboard

Advantages

- Saves time
- Better seat utilization
- Predicts rush hours
- Improves user experience
- Easy management

Novelty

Integration of:

- Machine Learning prediction
- Smart reservation
- Occupancy analytics

Requirements

Functional Requirements

- User Registration/Login
- View available seats
- Reserve seat
- Cancel reservation
- Admin dashboard
- Occupancy prediction

Non-Functional Requirements

- High availability
- Secure authentication
- Fast response time
- User-friendly interface
- Scalability

Hardware Requirements

- Computer/Laptop
- Wi-Fi

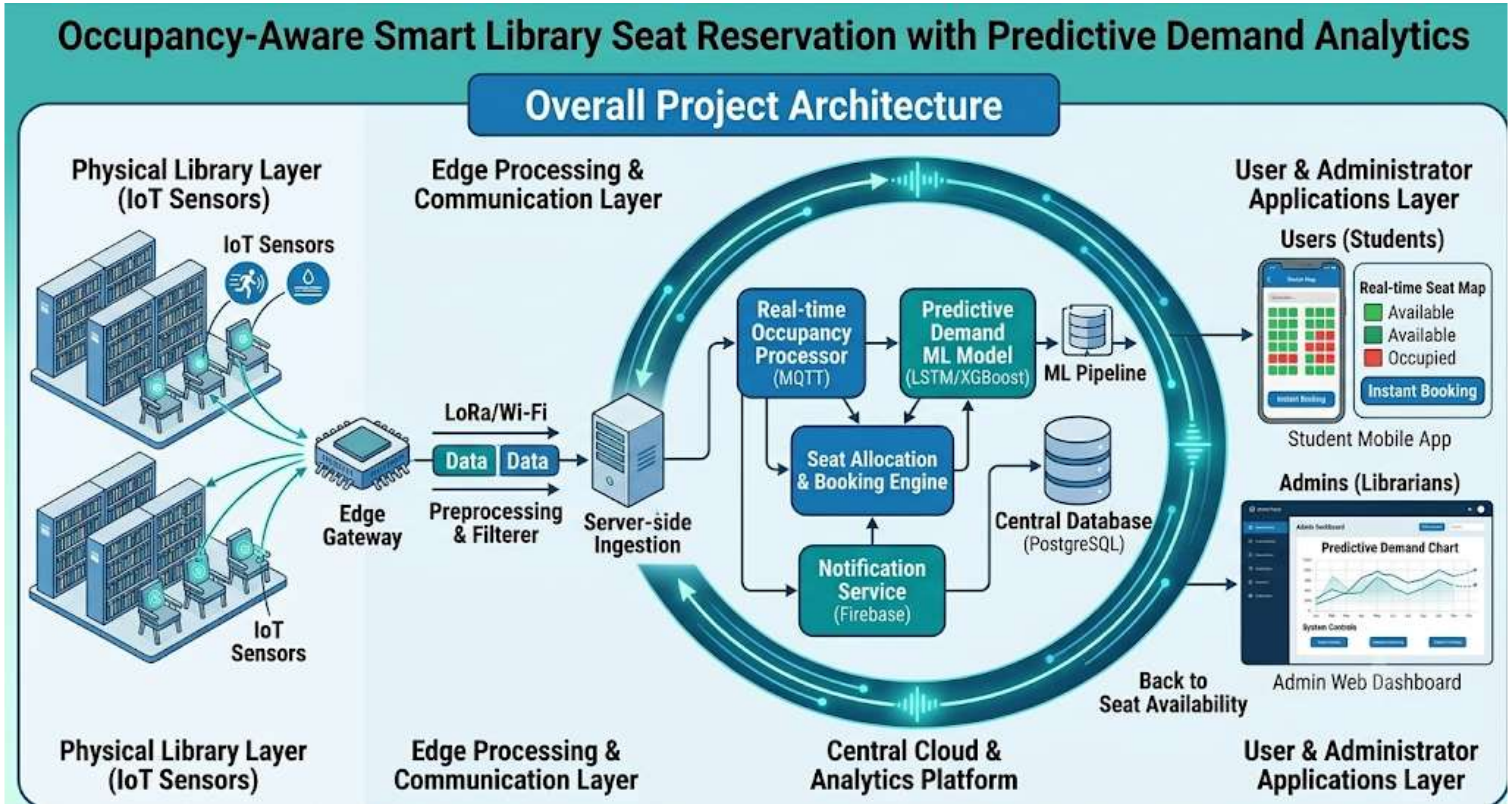
Software Requirements

- Python
- Flask/Django
- MySQL
- HTML/CSS
- JavaScript
- VS Code
- TensorFlow/Scikit-Learn

Dataset/APIs

- Library occupancy records
- Reservation logs
- Public occupancy datasets

System Architecture



Modules

- User Management
- Seat Reservation
- Occupancy Monitoring
- Prediction Engine
- Notification System
- Admin Dashboard

Planning

Work Breakdown Structure (WBS)

- Problem Identification
- Literature Survey
- Requirement Analysis
- System Design
- Database Design
- Frontend Development
- Backend Development
- ML Model Development
- Testing
- Documentation

Milestones

- Literature Survey Completed
- Requirement Analysis
- UI Design
- Database Completion
- ML Model
- Final Testing
- Documentation

Risk	Solution
Sensor failure	Backup manual update
Prediction errors	Retrain ML model
Server downtime	Cloud backup
Network issues	Offline caching

Expected Outcomes

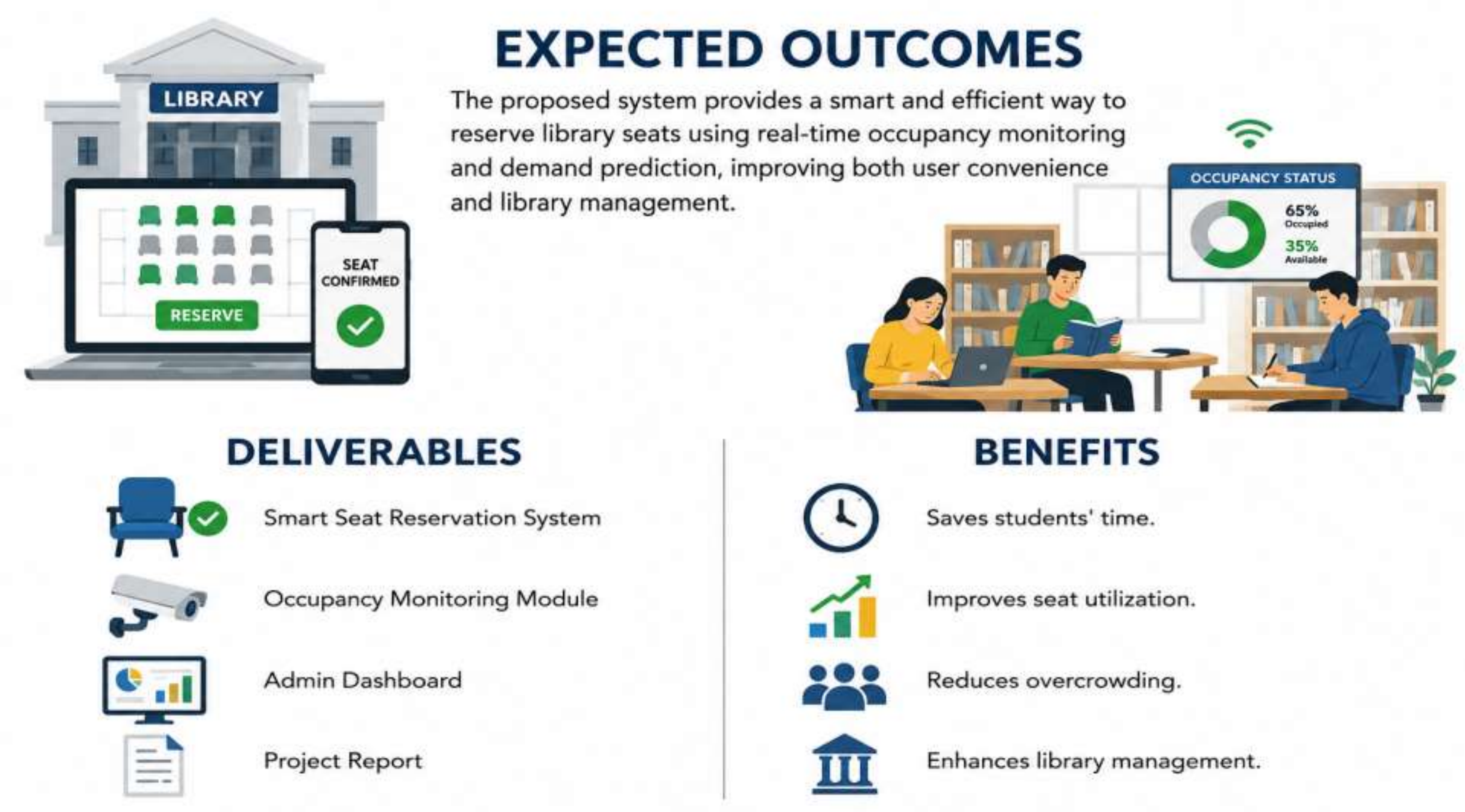
The proposed system provides a smart and efficient way to reserve library seats using real-time occupancy monitoring and demand prediction, improving both user convenience and library management.

Deliverables

- Smart Seat Reservation System
- Occupancy Monitoring Module
- Admin Dashboard
- Project Report

Benefits

- Saves students' time.
- Improves seat utilization.
- Reduces overcrowding.
- Enhances library management.



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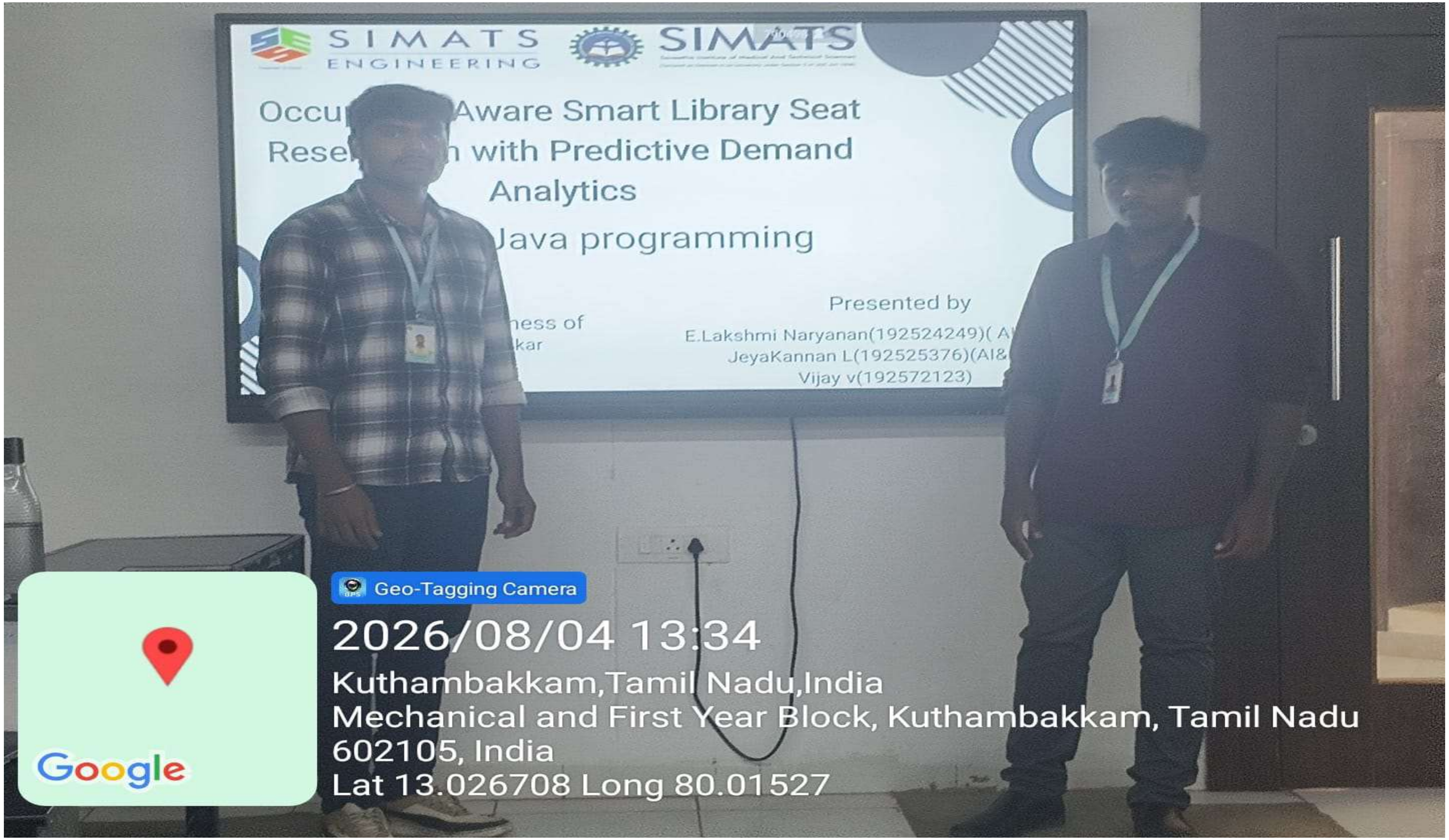
Thank You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